

SHIFT11

Theoretical Project & System Design

Revised architecture, event contract, analytics logic and validation strategy.

Design specification. Parameters and components described here are proposed until implemented and tested.

1. Problem statement

Traditional match statistics describe events but do not always reconstruct when a match changed, which players contributed, what sequence supports that conclusion, and why it mattered tactically. Shift11 connects event-level analytics, player attribution, evidence-chain reconstruction and grounded AI explanations.

2. Non-negotiable design principles

- Deterministic analytics establish what happened; AI explains why it mattered.
- Every shift, contributor score and tactical explanation must be traceable to validated events and reproducible calculations.
- Use synthetic data initially. Clearly label it synthetic; do not imply it is real Premier League tracking or event data.
- Separate measured evidence from inferred interpretation. Contribution scores are model estimates, not proof of exact causal responsibility.

3. Intended end-to-end architecture

Stage	Responsibility
1. Scenario configuration	Choose formation, seed, period settings and a controlled scenario.
2. Formation-aware generator	Create role-aware event candidates and coherent possession sequences.
3. Event validation	Reject invalid references, time ordering, coordinates and possession transitions.
4. Match state / spatial features	Compute territory, progression, pressure/recoveries, chance creation and possession control.
5. Momentum and shift detector	Calculate rolling momentum and apply persistence/supporting-event rules.
6. Player attribution / evidence	Score event contributions and reconstruct a chronological evidence chain.
7. Verified fact pack	Package metrics, contributors, event IDs, confidence/limitations and before/after comparison.
8. Microsoft AI layer	Use Agent Framework and Foundry to explain only the verified fact pack.
9. Dashboard	Show timeline, shift details, contributors, evidence and explanation.

4. Synthetic match generation contract

Formation and player roles

Initial formations: 4-3-3, 4-2-3-1, 4-4-2 and 3-5-2. Formation positions should influence plausible event selection, starting locations and support options. A role does not make a player perform only one action; it adjusts likelihoods and spatial expectations.

Connected possession sequences

- A possession has a single controlling team until an explicit transition occurs.
- Successful passes and carries usually preserve possession ID and establish an event-to-event chain.
- Failed passes, unsuccessful dribbles, tackles or interceptions may transfer control; recovery/turnover events must reflect that change.
- Shots, restarts and out-of-play outcomes need explicit rules. Do not select event team, player, event type and possession ID independently at random.
- Event timestamps must be ordered; event references must point to the correct match, team and player.

Shared pitch coordinate convention

Use a single absolute pitch frame: x and y range from 0 to 100 for the whole pitch. Define x=0 and x=100 as the two goal lines, and y=0 and y=100 as the two touchlines. Both teams use the same frame.

Choose one attacking direction per team for each half. For example, the home team may attack toward increasing x in the first half and decreasing x in the second; the away team does the opposite. At half-time, switch each team’s attacking direction. Convert event start/end locations into the shared pitch frame before comparing territory or progression.

This makes x-coordinates comparable across teams and periods. If a normalized attacking-relative feature is useful for analytics, derive it as a separate feature; do not mix it with stored absolute coordinates.

Controlled scenarios

Scenario	Expected event pattern to test	Important safeguard
High pressing	Pressure and recoveries near the opponent build-up; turnovers create transition opportunities.	A scenario label must not force a shift if calculated metrics do not support one.
Counterattack	Recovery/turnover followed by rapid forward progression and possible chance creation.	Events must form one chronological possession chain.
Defensive dominance	Repeated pressure, blocks/interceptions/recoveries and low-quality chances conceded.	Do not equate defensive actions alone with attacking momentum.
Harmless possession	Many low-risk passes with little progressive movement or chance creation.	Possession volume alone should not automatically create a major shift.

5. Event data contract

The current Python model includes team/player/match identifiers, timestamp, event type, x/y, optional end coordinates, pressure, speed, distance, outcome/success and possession ID. Keep the schema small but define each field consistently.

Field / rule	Proposed interpretation or validation
timestamp_seconds	Elapsed match time; non-negative and ordered. Period/half may be stored explicitly or derived unambiguously from time.
team_id / player_id	Must reference declared entities. Player must belong to the event team, unless an explicitly modelled exception is introduced.
event_type	Must belong to a documented taxonomy. Each type defines which endpoint, outcome and contextual fields are valid.
x, y; x_end, y_end	Absolute pitch coordinates in range 0-100. End coordinates are optional only for event types whose semantics do not require a destination.
success	Meaning is event-specific; document what success means for pass, carry, shot, tackle or interception rather than applying one generic meaning.
possession_id	Shared by events in the same possession; possession transition must be supported by an explicit outcome/action.
seed / scenario metadata	Saved with output to support reproduction and evaluation. Scenario metadata describes generation intent, not the measured result.

6. Validation and reproducibility tests

- Same seed + same configuration -> same event sequence and serialized output.
- All coordinates are within bounds; endpoints exist when required; no invalid IDs or negative timestamps.
- Event time is monotonic and the match duration/half-time boundary is consistent.
- Possession IDs do not silently jump between teams; transition events explain control changes.
- Formations contain the expected player roles and eleven-player starting groups for the synthetic baseline.
- Each controlled scenario is assessed by computed features and expected ranges, not by trusting the scenario label.

7. Initial momentum model

Proposed 0-100 team momentum is an aggregate of five normalized components. The weights below are initial design parameters and require validation against controlled scenarios and, later, appropriate external feedback.

Component	Initial weight	Possible signal family
Territory / field position	20%	Location of controlled actions and sustained play in advanced areas.
Possession value / progression	25%	Progressive passes/carries, line-breaking movement and controlled territory gain.
Pressure and recoveries	20%	Pressures, recoveries and defensive actions that regain or constrain control.
Chance creation	25%	Shots, entries and sequences that create meaningful goal threat.
Possession control	10%	Retention and sequence stability, without rewarding harmless circulation too much.

Window: five minutes, recalculated every 30 seconds, with recency weighting. Features must be normalized consistently and computed symmetrically for both teams. Momentum scores must remain bounded and missing features must have defined handling.

8. Initial shift rule

- Candidate momentum increase: at least 15 points.
- Change in relative momentum advantage: at least 10 points.
- Persistence: at least 60 seconds.
- Support: at least three meaningful supporting events.
- Hysteresis/cooldown: avoid duplicate markers for one continuing episode.

These thresholds are provisional. Evaluate false positives and missed shifts on controlled fixtures, then revise and document them. A detected shift is an analytical result, not an objective universal label for momentum.

9. Player attribution and evidence

Compute player contribution from verified event relevance, event quality/outcome, sequence position and the measured feature change. Return primary and secondary contributors, contribution scores, provisional tactical roles and event references. Avoid attributing all team-level movement to a single player without evidence.

Provisional roles: Momentum Catalyst, Progression Driver, Chance Creator, Defensive Trigger and Finisher. A chronological chain of about 5-7 events may show, for example, recovery -> progressive carry -> line-breaking pass -> final-third entry -> shot. Include metric changes before and after the candidate shift.

10. Verified fact pack and grounded AI

The fact pack is the boundary between deterministic analytics and generation. It should contain: match/scenario metadata; shift timestamp and team; before/after momentum and component metrics; trigger checks; event IDs with timestamps/types/outcomes/coordinates; ranked contributors and their supporting events; confidence or limitations; and the chosen audience mode.

- Microsoft Agent Framework is the orchestration option; Microsoft Foundry is the model/project/runtime path. Use the simplest supported integration that meets the demo requirements.
- The AI can summarize tactical significance or answer a question about supplied facts, but must not invent scores, shots, passes, player identities or causes.
- Where claims are material, the explanation should reference the supporting fact-pack items/event IDs. Missing evidence should lead to a qualified answer, not a guess.
- Personalized modes may change vocabulary and depth, but not the verified underlying facts.

11. Test strategy and minimum success criteria

Layer	Tests / acceptance
Generator	Seed determinism, formations, 90-minute bounds, coherent possession chains, plausible locations, half-time direction switch.
Validation	Invalid references, ranges, timelines, event-specific outcomes and possession changes are caught.
Analytics	Small fixtures yield expected component changes, rolling-window outputs and shift/no-shift outcomes.
Attribution	Known catalyst ranks appropriately; scores have documented meaning; every contribution references events.
Evidence/fact pack	Chronological chain, before/after metrics and all referenced IDs are valid.
AI	Adversarial/underspecified fact packs do not result in fabricated events or statistics.
End-to-end	Local run works from a clean setup; demo scenario is reproducible; cloud path is optional, not a blocker.

12. Conceptual entities (optional persistence)

The MVP can use JSON files. A relational database is optional until persistent multi-match history or queries are needed. Conceptual entities may include TEAM, PLAYER, MATCH, EVENT, MOMENTUM_STATE, MOMENTUM_SHIFT, PLAYER_CONTRIBUTION, EVIDENCE_EVENT and INSIGHT. This is a future design, not a claim that the tables exist.